

Content-Based Misinformation Detection on Short Claims and Full News Articles

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Abstract. Content-based misinformation detection is usually reported as a single accuracy figure on a benchmark of short claims. This article asks what that figure measures. On LIAR, three fine-tuned transformers and a frozen one are compared against a TF-IDF baseline under one protocol, and none separates from it: **the ceiling sits near 0.63 macro-F1 whatever the model**. That ceiling appears to belong to the claim format rather than to content analysis, since changing the unit of text from a one-sentence claim to a full news article reaches 0.805 and 0.735 on two corpora, a contrast that changes corpus as well as unit and is therefore a single comparison rather than a controlled one. Held-out topics cost between -0.004 and $+0.038$ macro-F1. Attention and leave-one-out occlusion, two attribution methods that fail in unrelated ways, agree on the same cues and share the same error. Carried onto 398 posts collected from Bluesky and labelled by the published credibility of the outlet they link to, every model lands between 0.503 and 0.590 where chance is 0.500. A classifier fitted on those posts instead reaches 0.889, which is an artefact: the platform renders the link inside the text, and holding outlets out by domain brings the figure to 0.547. A rule-based check-worthiness gate clears two posts in five, and audited against hand annotation it over-retains by 37%, so that proportion is an estimate with a wide margin. Read generously it still bounds where a content model can be asked anything.

Keywords: misinformation detection, transformers, explainability, distribution shift, dataset artefacts, check-worthiness.

1 Introduction

A false claim does not announce itself. The response the field converged on is to read the post and decide, and a large literature does exactly that, running from hand-engineered credibility features to fine-tuned transformers [6, 7, 8]. The premise is a sound one. What circulates on these platforms is carried by people and not by automation [1], and people act on what they read, so the wording is where the persuasion happens.

Two developments raise what the reading has to accomplish. A falsehood can be steered, which makes it a vector and not only a message, and puts the problem inside the security perimeter rather than beside it [2, 3]. Large language models have also lowered the cost of producing fluent misleading text, which can be harder to detect than text written by hand [4, 5].

The results reported for that line of work are usually single accuracy figures on a benchmark, and this article is about what such a figure actually measures. Our own starting point was unremarkable: fine-tune the standard encoders on LIAR [9] and report the best. The finding was that no transformer beat a TF-IDF baseline, which could be read as a failure of engineering. Pursuing it is what produced the substance of this paper. The ceiling turns out to belong to the shape of the data more than to the family of models, since changing the unit of text from a

claim to a whole article clears it by a wide margin. The same instinct, applied later to a sample we collected ourselves, exposed a classifier that looks excellent and is reading a URL.

Four questions organise the work, each stated below with the objective that answers it.

- **RQ1.** What accuracy can fine-tuned transformers reach on short political claims, and how much of it survives a change of topic?
RO1. Compare five content models under one protocol with paired bootstrap intervals, then retrain outside each of five topics and test on the held-out one.
- **RQ2.** What do their verdicts rest on, and can that be established by a method whose own weaknesses are known?
RO2. Attribute each verdict twice, with attention weights and with leave-one-out occlusion, two methods that fail in unrelated ways, and report only what both agree on.
- **RQ3.** How much of a real, collected stream of posts is even a legitimate target for a content model?
RO3. Gate a collected sample with explicit check-worthiness rules and audit that gate against hand annotation, so the proportion is reported with its error rather than as a measurement.
- **RQ4.** What does the credibility score this module produces contribute to a detector that reads propagation instead of text?
RO4. Hand the score to that detector at the cascade root and measure the change against a random control.

RQ4 is answered in Section 5, from the content side of the interface. The propagation model on the far side of it is the subject of a separate submission by the same authors, prepared from the same project and disclosed here in full.

This article makes five contributions.

- Five content models compared under one protocol, with the comparison tested: no gap between any model and a bag of words excludes zero under a paired bootstrap (Section 4.1).
- The ceiling those five share located in the claim format rather than in content analysis, by changing the unit of text from a one-sentence claim to a full news article (Section 4.2).
- The cost of a change of topic across five held-out topics for one model, which quantifies the shift without ranking the models under it (Section 4.3).
- Explanations from two methods with unrelated failure modes, reported for their agreement and for the error they share (Section 4.4), and an audit of the check-worthiness gate that decides which posts a content model may be asked about at all (Section 4.5).
- All five models carried onto a platform none was trained for, with a control that takes our strongest figure there from 0.889 to chance (Section 4.7), and an ablation of the resulting score inside a propagation model (Section 5).

2 Related work

2.1 From engineered features to pre-trained encoders

The first credibility classifiers for social media combined text features with user and citation behaviour [6]. Purely linguistic approaches followed, using n-grams, readability and psycholinguistic categories, and reported around 76 % accuracy across domains [7]. Those models remain relevant for two reasons: they are transparent, so a claim about what the classifier reads can be

checked, and they are the baseline any neural model has to beat. Our results suggest the second point is taken less seriously than it should be.

Pre-trained encoders changed the default. BERT [12] and RoBERTa [13] are the standard starting points for text classification, the second being the same architecture pre-trained longer, on more data, with better hyperparameters and without the next-sentence objective. The fine-tuning regime adopted here for a small dataset, a low learning rate over few epochs, is the one BERT’s authors prescribe for this case [12]. Recent surveys of misinformation detection place fine-tuned encoders at the centre of the content strand [8].

2.2 Verdict as one stage of a pipeline

Automated fact-checking is usually decomposed into four stages, check-worthiness, evidence retrieval, stance detection and verdict prediction, surveyed by Guo et al. [15]. Check-worthiness has its own literature and its own systems, ClaimBuster being the reference [16], evidence-based verification is served by FEVER [17], and stance detection by RumourEval [18]. The classifier studied here is the verdict stage operating without retrieved evidence, which is the common configuration in social media settings where no evidence base is available at posting time. We treat check-worthiness as a gate in front of it, and we report what that gate lets through on real data, because the answer bounds the whole module.

2.3 Explainability

A verdict without a reason is of little operational use, and the field has converged on local surrogate methods, LIME being the canonical one [19], and on attention as an accessible alternative. Attention is contested as an explanation, since where a model looks is not the same as what changed its answer, and dedicated architectures have been proposed to make the two coincide [20]. Our position is that neither method should be trusted alone and that agreement between two methods with different failure modes is the usable evidence, which is how we report it.

2.4 The known ceiling and its attributions

Surveys of the field agree that content-only detection is bounded, and attribute the bound to phenomena the text does not carry: misleading framing of true statements, manipulated context, and coordinated amplification [21, 22]. Our results are consistent with a bound but locate it more precisely. On short decontextualised claims the bound is near 0.63 and is reached by a bag of words. On full news articles it reaches 0.805 and 0.735. The limitation the surveys describe is real, but a substantial part of what is measured on the standard benchmark is the benchmark’s own format.

2.5 Benchmark shortcuts

Before asking how well a text classifier scores, it is worth asking which property of the text it has settled on. The literature supplies a warning at the extreme: Pelrine et al. matched the state of the art on Twitter16 without reading a single post, keying instead on the leading digits of the tweet identifier, a field that encodes when the message was written [10]. Where a benchmark can be solved without its text, a score obtained by reading that text is not self-evidently a measurement of reading. Fragility of the same kind appears once topics are separated across the split, which costs rumour detectors more than 40 % of their score and leaves them at the level of a trivial classifier [11]. Work collected under the name shortcut learning [23] treats these as one habit, a model settling on whatever correlates with the label most cheaply. Our own content classifier turns out to share it, and Section 4.7 reports the cue it found on the collected sample and what removing that cue costs.

2.6 Benchmarks and their platform

LIAR [9] and FakeNewsNet [24] are the community defaults for content-based work, and both are built on United States political fact-checking and Twitter-era engagement. Evaluation practice in the field has been criticised for relying on single splits of such resources [25]. That criticism motivated both our cross-topic protocol and the decision to collect a sample from a different platform. Bluesky serves that purpose here: its protocol exposes documented read endpoints without negotiated access, and the platform has itself been the subject of quantitative study [26].

3 Method

3.1 Data

LIAR [9] provides 12,800 statements fact-checked by PolitiFact with six veracity labels, in official splits of 10,240 training, 1,284 validation and 1,267 test items. We binarise the labels, mapping *pants-fire*, *false* and *barely-true* to misleading and the remainder to reliable, which is the usual reduction and makes the task comparable with the propagation side.

Full article text enters through UPFD [29], which stores one 768-dimensional encoding of each FakeNewsNet [24] news body at the root of the cascade that carried it. This supports one contrast and one only, the unit of text being judged. It is not the LIAR pipeline applied to longer inputs: the encoder is fixed rather than fine-tuned, the classifier is logistic regression over a precomputed vector, and the corpus and label provenance both change with the unit. Section 4.2 reports it on those terms.

Collected posts. A sample was collected from Bluesky, whose AT protocol exposes documented read endpoints and is therefore one of the few platforms on which such a collection is still possible without a commercial agreement.

We began where a content study naturally begins, by looking for posts that make misinformation-adjacent claims and reading them. Keyword retrieval returned 681, of which the check-worthiness gate kept 140. Those, plus 25 it had discarded, were annotated by hand. The gate that made that split is an earlier form of the rule set specified in Section 4.5, which sorts the same 165 records into 138 and 27.

What came back was not a small dataset but a lesson about the population. Only 93 of the 165 asserted anything a verdict could attach to at all, and of those 93, **87 were reliable and 6 misleading**. Accounts writing about these topics on this platform are overwhelmingly accounts refuting them, and a minority class of six supports no reportable metric. The gate had worked. There was almost nothing false to find by asking what posts say.

What worked was to stop asking the annotator and start asking the outlet. Posts are gathered by the *domain* their link points to, and each inherits the factual rating that the Iffy Index, built on Media Bias/Fact Check, publishes for that domain. The move buys scale and costs precision, and the cost is specific enough to name: the label now describes the source of what was shared and not the truth of what the post says about it. Three filters keep that substitution defensible. Only the two lowest factual tiers count, since the published file also carries roughly 1,300 domains rated *mixed*, whose inclusion would label every share of a mainstream outlet as misleading. A link preview must be attached, so that the domain is unambiguous. The link must be to an article. This yields **200 posts per class** across 34 domains, of which 398 carry text.

Two properties of this sample govern how it may be used, and both are stated wherever it appears below. The label is a property of the *linked outlet*, not a fact-check of the post, so an account refuting an unreliable article is labelled like one relaying it. The platform also renders the link as text inside the post, so the string the label was read from is present in the very field

a classifier reads, which Section 4.7 measures.

3.2 Three steps of increasing capacity

Models are compared under one protocol so that differences are attributable to one change at a time, as Figure 1 sets out. Differences quoted in the text are taken on the unrounded means throughout and can therefore fall 0.001 away from the subtraction of two rounded cells.

- **TF-IDF with logistic regression**, word and bigram features, minimum document frequency 2, at most 50,000 features, balanced class weights. This is the transparent baseline, and its contributions are exact: the effect of a term on a verdict is its TF-IDF weight times its coefficient, with no approximation.
- **Frozen DistilBERT** with mean-pooled embeddings and the same classifier on top. No weight is updated. This isolates what pre-training alone provides.
- **Fine-tuned BERT and RoBERTa**, all weights updated. RoBERTa is run in two regimes, a vanilla one and an optimised one, so that architecture and training regime are separated.

Figure 1 lays out the comparison, built so that one factor changes at a time, representation, then pre-training, then fine-tuning, so that every gain or its absence is attributable.

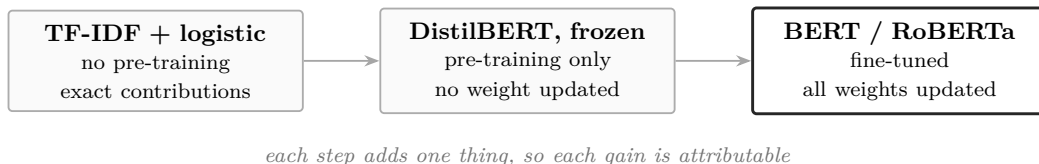


Figure 1. Three steps of increasing capacity.

The vanilla fine-tuning regime is AdamW, learning rate 2×10^{-5} , batch size 32, three epochs, maximum sequence length 64 tokens, no warmup and no weight decay, with the checkpoint selected on validation macro-F1 and the test split used once. The optimised regime differs in four settings and nothing else: four epochs, a warmup over the first 10 % of steps, weight decay 0.01 and a maximum length of 128 tokens. Implementation uses PyTorch [27] and the Transformers library [14]. Seeds are fixed and reported experiments are reproducible from a single command.

3.3 Check-worthiness as a gate

Not every post states something a verdict can be about. We place an explicit rule set in front of the classifier, after ClaimBuster in intent [16] but not in method: markers for quantities, attribution verbs, institutional subjects, change and measurement raise a score, and first-person markers, opinion vocabulary and interrogatives lower it, with a small bonus for length. Items scoring at least 0.3 are retained.

Rules rather than a model, for a stated reason: no labelled check-worthiness corpus exists for this platform and this domain, and a rule can be read and contested where a model trained on nothing would be a black box. The stage is also self-scoring, because the annotation tool records a skip whenever an item turns out not to be a claim, which lets the gate be audited.

3.4 Two attribution methods

For the linear model, contributions are exact. For the transformers we use two methods and report their agreement.

Attention is read from the final layer, averaged over heads, on the row the classifier consumes. It shows where the model looked, which is not the same as what changed its answer, and we treat it as such.

Leave-one-out occlusion removes one token at a time and measures the signed shift in the output probability. It shows what mattered, but it ignores interactions between tokens, so a cue that only works in combination is invisible to it.

The two have disjoint blind spots. Where they agree, the agreement is evidence. Where they diverge, neither is taken as an explanation. A case where they share an error is reported alongside, since agreement and shared failure are the same evidence read in two directions.

3.5 Cross-topic and cross-platform protocols

Topic shift. Five topics are identified in LIAR by keyword. For each, the model is trained on the remaining four and evaluated on the held-out one, against an in-domain control trained on the same number of items. The reported quantity is the gap between the two.

Platform shift. Five models are applied unchanged to the collected posts: the four fine-tuned transformers, including the weight-decay run held outside the protocol of Table 1, and the bag of words. The frozen encoder is not among them: its predictions on the collected posts were not computed, which leaves the comparison below without its middle rung. Because the label semantics differ between the two corpora, the transfer figure alone would not be interpretable, so we additionally report the distribution of the answers and we fit a classifier on the collected posts directly. That last step requires a control, described with its result in Section 4.7, because the platform writes the link into the post and the label is read from that link.

4 Results

4.1 Five models, one protocol

Table 1 reports the binary reliable against misleading task on the test split. Intervals are 5,000 bootstrap resamples of the 1,267-item split, and they overlap completely, which is the result.

Model	Accuracy	Macro-F1	95 % interval
TF-IDF + logistic regression	0.637	0.633	[0.607, 0.658]
DistilBERT (frozen) + LR	0.631	0.627	[0.600, 0.653]
BERT (fine-tuned)	0.627	0.616	[0.588, 0.643]
RoBERTa (vanilla fine-tune)	0.646	0.632	[0.604, 0.658]
RoBERTa (optimised fine-tune)	0.636	0.624	[0.597, 0.649]

Table 1. Content detection on LIAR, five models under one protocol.

Table 1 and Figure 2 report the comparison. Three observations follow.

In Figure 2 the dashed line is that baseline. Neither pre-training alone, nor fine-tuning, nor a tuned regime moves the figure, and no gap against the baseline excludes zero under a paired bootstrap. On a split of this size that establishes that we cannot tell these models apart, which is weaker than establishing that they are equivalent and is all the argument requires.

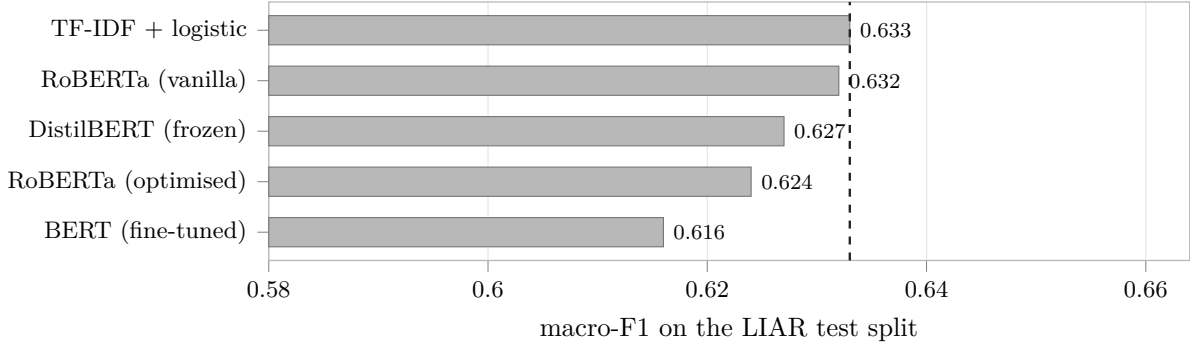


Figure 2. No transformer separates from the TF-IDF baseline.

RoBERTa beats BERT, which is the expected ordering, though the pair differs in pre-training and in configuration at once and this comparison separates neither. The optimised regime shows no detectable gain over the vanilla one. No transformer separates from the bag of words, TF-IDF standing at 0.633 and the best transformer at 0.632.

Separation is the word to use, not ranking, because on a 1,267-item split a ranking of figures three decimals apart is not a finding. The split is resampled 5,000 times and both models are rescored on each draw, so that the sample’s own variance cancels. No transformer’s gap against TF-IDF excludes zero. The vanilla fine-tune sits at -0.001 with an interval of $[-0.031, +0.029]$, and the widest deficit, BERT’s, at -0.016 with $[-0.044, +0.012]$. The frozen encoder sits at -0.006 with $[-0.034, +0.022]$. A sixth run held outside this protocol, RoBERTa with the optimiser’s default weight decay left in place, scores 0.647 and is reported in Table 3 for completeness. It is the one figure here above the baseline, and it does not survive the same test either, its gap being $+0.015$ with an interval of $[-0.014, +0.045]$. The ceiling is therefore not an artefact of which run we chose to headline, and Section 6.1 sets it beside an independent study of the same benchmark.

Figure 3 shows five of the six models of Table 1 reading one claim, the frozen encoder excepted because its weights were not kept. Above, the probability each assigns is drawn outward from the 0.5 boundary, dark towards *reliable* and amber towards *misleading*. Three answer one way and two the other, and they span 0.300 on an input where the held-out benchmark separates the same five by 0.031. Below are the four tokens that moved each verdict most, measured by deleting one token at a time and normalised so the largest is 1.00. The word *officials* carries $+1.00$ for three of the models and -1.00 for BERT, so five models a benchmark cannot separate do not agree on which words carry the verdict.

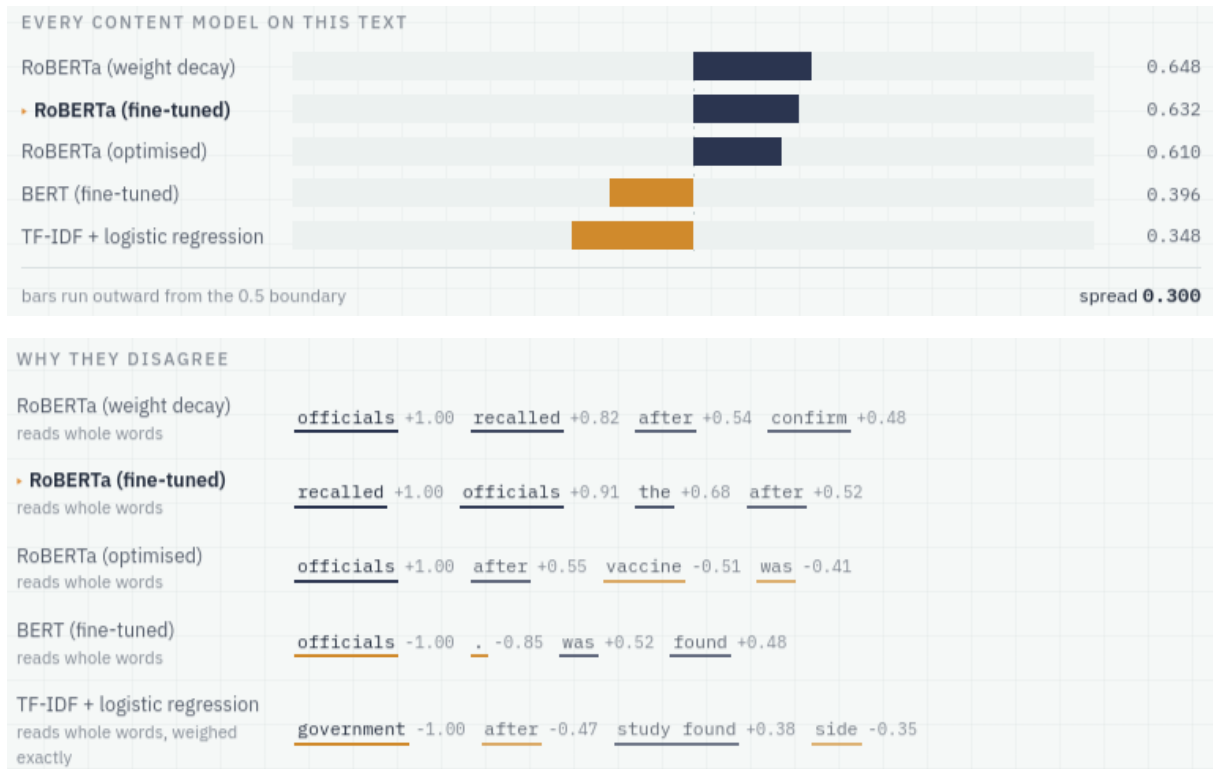


Figure 3. A three-point spread on a single sentence.

The frozen DistilBERT row is the informative one. If pre-trained representations carried usable signal for this task, freezing them would still capture part of it and fine-tuning would extend it, and neither shows up above the noise of this split. Section 4.2 supplies the positive evidence, by changing the unit of text and watching the ceiling move, and Section 6.2 reads the two together.

4.2 The ceiling and the claim format

The obvious objection to the previous section is that it indicts content analysis in general. It does not, and the contrast that shows this is a single change of unit.

UPFD [29] builds each propagation cascade as a tree whose root is the news item and whose remaining nodes are the accounts that shared it. Under its *bert* feature set the root carries a 768-dimensional embedding of the full news article, drawn from FakeNewsNet [24], and each account carries an embedding of its own posting history. Logistic regression on the root vector alone, reading no account and no edge, reaches **0.805** macro-F1 on PolitiFact, with a bootstrap interval of [0.749, 0.855], and **0.735** on GossipCop, with [0.721, 0.749], against the ~ 0.63 that every model in Table 1 reaches on LIAR’s single sentences. Both fits use the published train split, balanced class weights and no other tuning. A body of text carries structure, source attribution, hedging and internal consistency, none of which survive compression to one sentence stripped of its context.

The comparison is not perfectly controlled. The representation changes, the corpus changes, and the labels come from a different provenance. What it does establish is the negative claim the argument needs, that the ceiling is not a fixed property of reading text for credibility, since the same broad approach applied to a longer unit clears it by a wide margin on both corpora.

On the evidence here the 0.63 bounds what a short decontextualised claim contains rather than what a content model can do, though the contrast that shows it moves corpus and unit together. The gap is roughly seventeen points of macro-F1 on PolitiFact and ten on GossipCop, and

Section 6.2 places it against what the surveys of this area attribute the bound to.

4.3 Topic shift

Table 2 trains outside a topic and tests on it, in macro-F1, with gaps taken as differences of the printed cells. The per-topic figures for the fine-tuned RoBERTa are not available, only its mean gap, so no ordering between the two models is claimed.

Held-out topic	In domain	Cross-topic	Gap
health	0.629	0.608	+0.021
economy	0.603	0.607	−0.004
immigration	0.615	0.577	+0.038
elections	0.627	0.591	+0.036
taxes	0.608	0.595	+0.013

Table 2. Generalisation under topic shift, for the TF-IDF baseline.

In Figure 4 the held-out bar is shorter almost everywhere, by up to 0.038 on immigration and 0.036 on elections, and taller on economy, whose items resemble the rest of the corpus closely enough that holding them out barely shifts the training distribution.

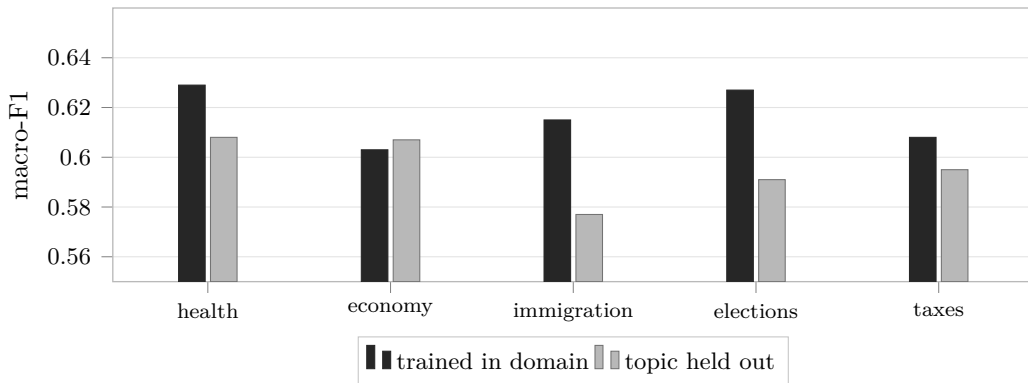


Figure 4. The cost of a change of topic.

What a change of subject matter costs is reported in Table 2 and Figure 4. The drop is real, modest and topic-dependent, from -0.004 on economy to $+0.038$ on immigration, with elections close behind at $+0.036$. The negative gap is not an anomaly to be explained away: economy items resemble the rest of the corpus closely enough that holding them out barely shifts the training distribution. The gap runs from -0.004 to $+0.038$ across the five topics, so the cost reaches four points at its largest and reverses sign on one topic.

The ordering of models under shift is the question a deployment would ask, and on five topics we cannot answer it. The fine-tuned RoBERTa carries a mean gap of $+0.014$ against the baseline’s $+0.021$, a difference of 0.007 across five held-out topics with no dispersion reported, which is not a ranking. These five topics are measured for one model and no dispersion is available for either, so **they quantify the size of the shift and do not rank the two models**. For a property on which the field routinely ranks models, that is worth stating, and Section 6.3 says whose criticism it restates.

4.4 Evidence behind the verdicts

Figure 5 reads the same sentence with the fine-tuned RoBERTa and scores it twice, captured from the system of Section 5, which states what each panel measures. Darker means larger, and on the right an amber underline marks a token that pushed the verdict towards *misleading*. What matters is the agreement: two methods that fail in unrelated ways settle on the same two cues, *officials* and *recalled*, and neither bears on whether the claim is true. The model has learned which register a checked claim is written in.

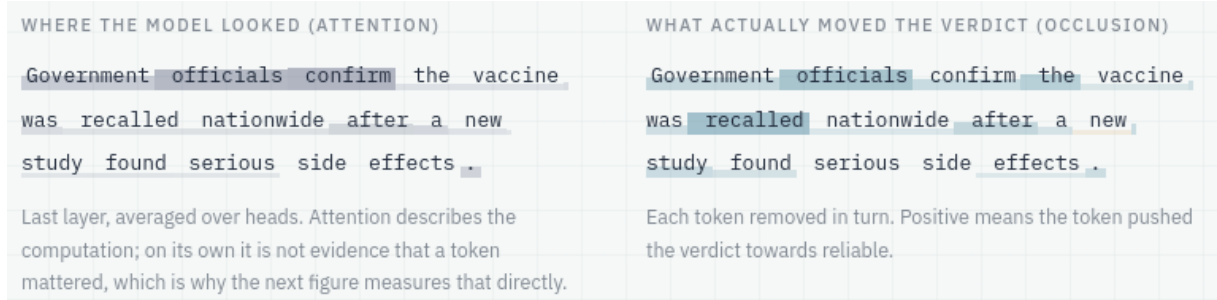


Figure 5. Two attributions over one claim.

For the linear model the contributions are exact. The strongest cues toward *misleading* are partisan and attack vocabulary, including named political figures and *government*, and the strongest toward *reliable* are verifiable quantities, *percent*, *million*, *than*, *more*, *average*. The model has learned a register rather than a fact-checking procedure, which is exactly what the 0.63 ceiling would predict.

The fine-tuned transformer agrees. Explained independently by attention and by occlusion, it leans reliable on *said*, *voted*, *less*, *more*, *half* and on numerals, and misleading on *nothing*, *government* and *health*. The two methods have disjoint failure modes, so their agreement on the same cues is evidence in a way that either alone would not be.

They also share an error, and reporting it matters more than reporting the agreement. Both attribute weight to *will*, a future auxiliary that carries no veracity information and is simply frequent on the misleading side of this corpus. Two methods pointing at the same spurious cue is a reminder that agreement establishes what the model uses, and says nothing about whether it is sound. Section 6.4 draws out what that implies for how attributions should be reported.

4.5 Check-worthy share of a real stream

Two audits of the check-worthiness gate are available, on different populations, and they should not be conflated.

On the 165 records that were individually annotated, the gate can be scored against the annotator’s own skips. Of the 138 it retains, 37% were skipped as non-claims, and of the 27 it rejects, 78% were indeed non-claims. It sorts in the right direction while letting a substantial amount of noise through. Crossed against the verdicts, the gate retains 81 reliable claims, all 6 misleading ones and 51 non-claims, and the 6 claims it loses are reliable without exception. The two counts of 87 and 6 that fall out of this table are the retained claims and the reliable claims on one side, the lost claims and the misleading claims on the other, and they coincide by arithmetic rather than by identity: 81 of the 87 belong to both.

This audit is optimistic by construction and should be read as such. Those 165 records were themselves selected for annotation by an earlier form of the same rule set, so posts the present rules would keep and the earlier ones dropped are represented only by the 25 rejected records that were annotated as a check. Scoring a filter on a sample the filter helped choose bounds

what the figure can establish. Its six false negatives share one shape: short breaking-news lines and quoted exchanges, carrying no number, no attribution verb and no institutional vocabulary. A claim stated in four plain words is invisible to a rule set over surface markers, which is the expected failure mode and more useful to name than a single accuracy figure.

On the 398 source-labelled posts, the gate measures reach rather than correctness. **168 of 398 clear the floor**, a mean score of 0.224. Four posts in seven clear nothing the gate recognises as a claim, which on a rule whose own error is characterised above should be read as an order of magnitude and not a measurement. What the gate keeps out are reactions, commentary and bare headline fragments accompanying a link.

That is not a defect of the sample, which was assembled to study what accounts circulate rather than what they assert. The rate differs by class, 36.2% on the misleading side against 48.2% on the reliable one, consistent with mainstream outlets being shared by quoting an attributed headline and partisan ones by commentary. Section 6.5 takes up what the proportion bounds.

4.6 Transfer off the training platform

Table 3 applies the LIAR-trained models unchanged to the 398 collected posts, in macro-F1, where chance is 0.500 on this balanced sample. The last row is the run held outside the protocol of Table 1, listed for completeness.

Model	LIAR	Collected	95 % interval	Answered “reliable”
RoBERTa (fine-tuned)	0.632	0.558	[0.506, 0.607]	16.3 %
RoBERTa (optimised)	0.624	0.509	[0.460, 0.557]	17.6 %
BERT (fine-tuned)	0.616	0.503	[0.454, 0.552]	16.1 %
TF-IDF + logistic regression	0.633	0.590	[0.538, 0.636]	33.7 %
RoBERTa (weight decay)	0.647	0.565	[0.514, 0.614]	16.8 %

Table 3. Transfer to the collected platform.

The preamble matters. A LIAR label is a fact-checker’s verdict on the claim a sentence makes. A label here is the published factual rating of the outlet the post links to. Carrying a model across therefore does not ask whether it is still accurate, it asks a model trained to judge claims to predict a property of a domain name. A poor figure is expected, and reporting it alone would say nothing.

What the figures do say is in the last two columns of Table 3. Two of the five, the optimised RoBERTa and BERT, have intervals straddling 0.500 and are not distinguishable from a coin. The other three clear chance, the bag of words by the widest margin at 0.590, with an interval opening at 0.538. Every model loses ground against its own LIAR figure, from 0.043 for the bag of words to 0.115 for the optimised RoBERTa. The manner of the failure is more telling than its size, and it is that they get there by answering *misleading* for more than eight posts in ten, TF-IDF for two in three. This is a collapse onto one class, not a graceful degradation.

Of the five transferred, the simplest transfers best. TF-IDF, which was not beaten on LIAR either, is the only one whose answers remain distributed across both classes. Section 6.6 weighs the reading that invites and says why the sample had to come from outside the standard resources.

4.7 Leakage control on the collected sample

Table 4 reports what the collected labels are predictable from in the text, as the mean of three seeds with deviations of 0.042, 0.028, 0.046 and 0.057 from the top row down. The domain split holds a third of the outlets out entirely, and its two rows are within a deviation of each other

and of chance.

Split	Text	Macro-F1
by post	as posted	0.889 ± 0.042
by post	links removed	0.700 ± 0.028
by domain	as posted	0.547 ± 0.046
by domain	links removed	0.542 ± 0.057

Table 4. The link, not the claim.

In Figure 6 the leftmost bar is what a classifier fitted on these posts reports if nothing is controlled. Removing the link costs nineteen points, holding unseen outlets out brings it to chance, and removing links on top of that changes little, the score having already fallen to chance.

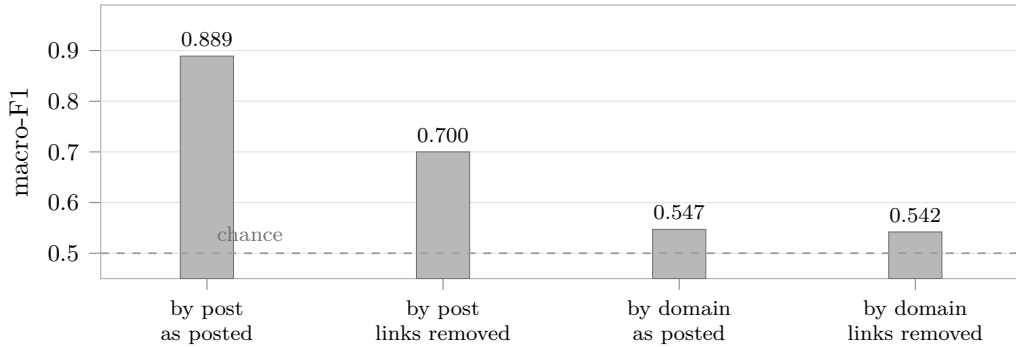


Figure 6. An apparent result and what survives control.

A classifier fitted and evaluated on the collected posts reaches **0.889**, which read on its own would be the strongest content result in this article. It is an artefact.

The platform renders a link as text inside the post, and the label is read from precisely that link, so a model can score well by learning which domains carry which label, rather than anything about the claim. Table 4 and Figure 6 cross the split with the removal of that string. Removing the link costs nineteen points. Holding unseen outlets out brings the figure to **0.547**, which on a balanced binary task is chance, and removing links on top of that changes little further, the score having already fallen to chance.

Neither control is ours, and Section 6.7 places both against the studies they come from.

One consequence is practical and belongs here. The anonymised release accompanying this work strips links explicitly rather than incidentally, which was not true of the first export. The pattern that pseudonymises account handles matched the commonest link endings, leaving those on other suffixes in place.

5 The content score as an input to a propagation model

The classifier described so far is one of two sub-modules of a larger system. The other reads propagation structure rather than text and is the subject of a companion paper. This section reports what our output is worth on the other side of the interface, from this side of it.

Figure 7 takes one collected cascade of 581 accounts. Our classifier reads the wording and answers *misleading*, 0.353, which on this story is the wrong side of the boundary. The propagation model

reads the cascade and answers *reliable*, 0.603. Handing our score over and letting that model decide again gives 0.628. A score that is weak, and here pointing the wrong way, still moves its consumer by +0.026, because what crosses the interface is a feature and not a verdict. This bounds what a content module should be judged on: its accuracy as a standalone classifier is not the same property as its worth to a system that reads something else, and Table 5 measures that worth over a whole split.

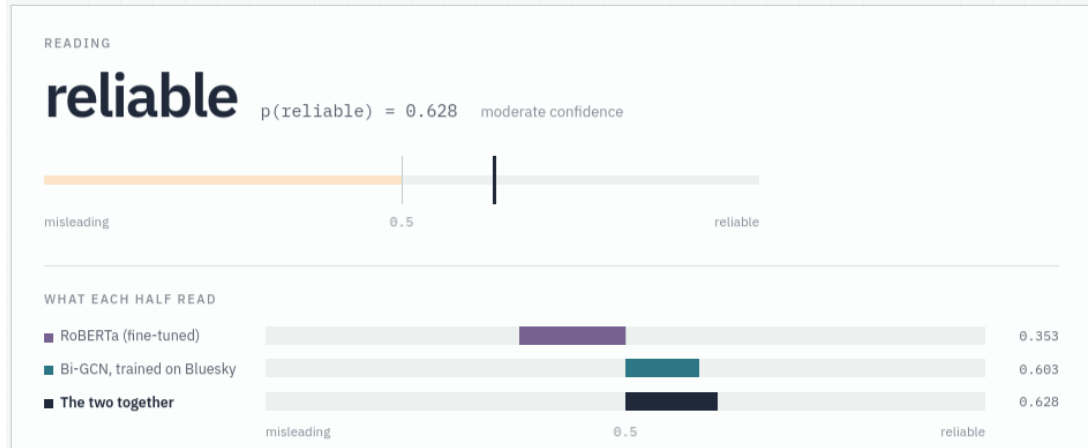


Figure 7. One cascade, read by both halves of the system.

5.1 The interface

What leaves this module is a single number per story. It is handed over as one extra feature column on the root of the cascade, and the receiving model is trained separately, on its own loss, sharing no representation with ours.

Handing over so little is a choice about measurement. Everything this module contributes is concentrated in one scalar, so the question of what it contributes has an arithmetic answer: run the receiver with our column, with a column of random numbers, and with no column, and read off the difference. Had the two models been fused, as recent hybrid work favours [30, 31], the combined system would probably score higher and this section could not have been written.

The number handed over is produced by a small multilayer perceptron over the news article embedding the receiving benchmark supplies. It reads the same input as the article-level classifier of Section 4.2 and not the LIAR model of Section 4.1, which was trained on another corpus at another unit of text. Fitted on the published training split and measured on the held-out one, it reads 0.823 ± 0.004 on PolitiFact against the 0.805 of Section 4.2, the difference being the classifier and not the input, and 0.735 ± 0.000 on GossipCop, where the two agree.

One precaution is required for that arithmetic. Our scores are cross-fitted over five folds, so no story is ever scored by a version of this module that saw it during training. Skip that and the column carries the answer, and any gain measured downstream is our own label handed back to us.

5.2 Contribution of the content score

Table 5 reports the ablation as the mean of three seeds, with a per-cell standard deviation between 0.004 and 0.017, so the +0.056 of the first row clears its own noise and the +0.003 of the second does not. The experiment belongs to the companion propagation paper, which specifies it in full, and is reproduced here because it carries this paper’s only measurement of the content score inside a working system. The *profile* node features are ten account counters,

and the *bert* features are a 768-dimensional embedding of each account’s posting history, with the news article at the root.

Dataset	Node features	Graph only	+ random	+ content
PolitiFact	profile	0.787	0.783	0.843
PolitiFact	bert	0.828	0.828	0.831
GossipCop	profile	0.909	0.909	0.905
GossipCop	bert	0.914	0.935	0.929

Table 5. What the content score adds to a propagation model.

The first row of Table 5 is the result. On PolitiFact, where the cascade’s nodes carry only account counters and the graph model has no access to text of any kind, the content score raises macro-F1 from 0.787 to **0.843**, which is a quarter of the error that remained. The random control moves nothing, so the gain is the signal and not the extra dimension.

The second row qualifies it. Where the nodes already embed the text of each account’s posting history, the same score adds +0.003. The content signal is not additive in general. **Its value is conditional on what the other model can already see.** A system that reported only the first row would be claiming a general result from a specific one.

The fourth row disciplines the other three. There the *random* control gains +0.021, where the real score gains −0.004 on the row above it and +0.003 on PolitiFact *bert*, which places the noise floor of this experiment above the effect being measured in that block. No conclusion is drawn from that block.

6 Discussion

The sections above are measurements. This one reads them, against the studies whose methods and tests they reuse and against what a system built on this module would have to assume. Table 6 sets out the first of those comparisons in one place: each study, the method or test taken from it, and what that yields here, with our own result on the last row.

Study	Method or test reused	Outcome here
Wang [9]	The LIAR benchmark and its binary protocol	A ceiling near 0.63 macro-F1 shared by five models
Devlin et al. [12]	A pre-trained encoder fine-tuned on the task	0.616, below the lexical baseline
Liu et al. [13]	The same with a longer pre-training budget	0.632, the best of the five and still not separable
Hassan et al. [16]	Check-worthiness as a stage before the verdict	Two posts in five clear the gate, which over-retains by 37 %
Ribeiro et al. [19]	Perturbation-based attribution	Occlusion and attention settle on the same cues, neither bearing on truth
Pelrine et al. [10]	A simple predictor run as a control on a reported result	0.889 on collected posts, and 0.547 once the link is removed
Wu and Hooi [11]	Event-separated evaluation against spurious cues	Holding outlets out by domain brings the classifier to chance

Study	Method or test reused	Outcome here
This article	Five models under one protocol, every margin tested by paired bootstrap	No gap against the TF-IDF baseline excludes zero , and the ceiling moves with the unit of text, not with the model

Table 6. Prior work whose method or test this article reuses.

6.1 What the benchmark can and cannot separate

The ordering of the second and third rows of the table, which is the ordering a benchmark paper would report, is not a result on this split. The five models span 0.031 macro-F1 while the bootstrap interval on any pairwise gap contains zero, so what the literature records as an improvement from a longer pre-training budget is, at this sample size, indistinguishable from noise. The finding is not that these models are equivalent, which our data cannot establish, but that this benchmark cannot tell them apart.

Two external readings support that. The best transformer sits at 0.632 against 0.633 for the bag of words, which is close to the range classical linguistic features were already reported to reach on this kind of task [7], so the pre-trained encoders are not buying what a decade of feature engineering had not already bought. And an independent 2025 study of LIAR reports the same hard ceiling and the same failure of transformers to separate from simple linear models [28], which places the ceiling in the benchmark rather than in our protocol.

The single-sentence panel adds a caution about how such figures should be read. Five models a benchmark cannot separate span 0.300 on one claim and disagree on which words carry the verdict, so models that a benchmark cannot tell apart are not reading the same evidence, and a single accuracy figure conceals that.

6.2 Where the ceiling comes from

Surveys attribute the bound on content-only detection to manipulated context and coordinated amplification [21, 22]. Our contrast suggests a more mundane component sits alongside those, namely the unit of text the benchmark is built from [9]. Reading the same broad approach on a full article clears the ceiling by roughly seventeen points of macro-F1 on PolitiFact and ten on GossipCop, and the frozen-encoder row points the same way from the other side, since a representation that carried usable signal would show part of it under freezing.

The consequence for the literature is a comparison problem rather than a modelling one. Any comparison drawn between systems evaluated on LIAR and systems evaluated on article-level corpora is comparing tasks, and the difference is large enough to swamp most reported model improvements. A survey table that lists both kinds of figure in one column is not measuring one thing.

6.3 Robustness under topic shift

The gap we measure across five held-out topics quantifies the size of a shift without ranking the models under it, because the per-topic figures exist for one model only. Reporting that limit rather than the mean gap is the concrete form, in our setting, of the criticism of single-split evaluation [25]. Robustness to a change of subject matter is a property the field routinely ranks models on, and five topics measured for one model do not support such a ranking.

6.4 What the attributions establish

Two attribution methods with disjoint failure modes agreeing on the same cues is evidence in a way that either alone would not be, and their shared error on *will* is the more useful half of the finding. It follows that agreement establishes what a model uses and says nothing about whether what it uses is sound. That is the practical case for not relying on a single attribution method [19], and part of the motivation for architectures that make the explanation part of the prediction [20]. It also fits the ceiling: a model that has learned which register a checked claim is written in would attribute exactly as this one does.

6.5 Check-worthiness and the case for a second signal

Four in seven collected posts clear nothing our gate recognises as a claim, on a rule whose own error we characterise rather than hide. Read generously the proportion still bounds where a content model can be asked anything, and in bounding this module it supports the two-signal design: where the text asserts nothing checkable, a verdict has to come from somewhere else, which is the propagation signal. The argument for reading structure alongside text is therefore empirical here rather than architectural, and it rests on a property of real streams that a benchmark of curated claims cannot exhibit.

6.6 Off the training platform

The collapse onto one class could not have been observed on the standard resources, which share a platform and an era [9, 24], which is why the sample was collected from a platform outside that ecosystem [26]. That the simplest model transfers best invites the reading that transfer degrades with model complexity, and these five points cannot support it: the configuration that would sit between the bag of words and the fine-tuned transformers, the frozen encoder, is the one missing from the table. What the figures do establish is narrower and still useful, that a bag of words has less to overfit to the idiom of a single corpus.

6.7 A shortcut in a sample built to avoid them

Neither control we ran is novel. A trivial feature outscoring a trained model is what Pelrine et al. documented on Twitter16, where the first three digits of a tweet identifier reached the state of the art [10], and holding outlets out is Wu and Hooi’s event-separated protocol [11] with outlets in the role of events. What is new is the shortcut itself, a link rendered into the text of the very post whose label was read from that link, and the fact that a sample collected specifically to escape the benchmarks turned out to carry one of its own.

The reuse consequence follows from it. The collected sample is a good object for studying who circulates what and an unreliable one for any text classifier fitted on it without holding outlets out, so we recommend the domain-grouped protocol for any reuse of it.

6.8 Implications for the content signal

Read from the content side, the ablation says something the LIAR results cannot. A content classifier reading the article, at 0.823 on this corpus, is worth a quarter of the remaining error to a model that cannot read text at all. The ceiling of Section 4.1 is a property of the claim format and not of this module’s utility within a system, and the two figures are not interchangeable: what the receiver gains here is what an article-level reader supplies, not what a short-claim reader supplies.

It also delimits that utility. The gain is visible only where the consumer is blind to language, and the one configuration where the consumer already reads text shows no gain the random control does not match. Combined with the finding of Section 4.5, that most real collected posts state

nothing checkable, the picture is consistent: the content module is a narrow instrument with a real but bounded contribution, and the case for reading propagation alongside it is empirical rather than architectural.

One scope note bounds this section. The integration experiment is computed on benchmark cascades throughout, and no score derived from collected text enters a graph model anywhere in this work. The artefact of Section 4.7, where a classifier fitted on collected posts read the link rather than the claim, therefore cannot have reached the figures above.

7 Conclusion and limitations

Content-based misinformation detection on short claims stops near 0.63 macro-F1, and it does so for every model we tried, including a bag of words. Ceiling is the accurate word and ranking is not. Resampling the 1,267-item test split 5,000 times, no transformer’s gap against TF-IDF excludes zero, including a sixth run that scores above the baseline and still fails the test. That ceiling appears to be a property of the claim format rather than of content analysis, since a classifier over the full news article reaches 0.805 and 0.735 on two corpora, on a contrast that changes corpus alongside unit. Held-out topics cost between -0.004 and $+0.038$ on one model, which quantifies the shift without ranking the models under it. Explanations from two methods with different failure modes agree on a register rather than on a fact-checking procedure, and share one spurious cue. On a platform the models were not trained for, all of them land between 0.503 and 0.590, where chance is 0.500, and two of the five are indistinguishable from a coin. They reach those figures by answering misleading for more than eight posts in ten, the bag of words for two in three. A classifier fitted on that platform instead reaches 0.889 by reading the link rather than the claim. Read from the other side of the interface, a classifier reading the full article, at 0.823 on that corpus, is worth a quarter of the remaining error to a detector reading account counters on PolitiFact, and nothing on the three other configurations, so its value is conditional on its consumer and on the corpus.

Limitations. LIAR is single-sentence, United States political and fact-checked by one organisation, so the ceiling we locate is established on one corpus, and the article-text contrast changes representation and corpus at the same time, which makes it a single comparison rather than a controlled one. The collected sample is labelled by outlet rather than by claim, so an account refuting an unreliable article is scored like one relaying it, the label being a property of what was shared. The check-worthiness rate rests on a rule set whose own error is characterised but not eliminated, and the annotated records that characterise it were themselves selected by an earlier form of that rule set, which makes the audit optimistic by construction. The domain-grouped condition rests on eleven held-out outlets per fold. The explanations concern one model on one corpus and are not claims about transformers in general. Nothing here is offered as a causal statement, every figure being an association between a text and a label assigned to it.

Data and code availability. The 398 posts this article analyses, with account identities pseudonymised, the code for every experiment and the fitted models the tables report are available from the authors on request, and will be released publicly on publication. Outlet links are stripped from the released text, which matters here more than it would elsewhere. Those links were the cue our own classifier learned in Section 4.7, and publishing them beside the labels would hand the next reader the same artefact. Every figure is reproducible from one command at a fixed seed.

Declaration of AI use. During the preparation of this work, the authors used Claude (Anthropic), a large language model, to assist with writing and debugging the code for the experiments and the data processing, and with drafting and editing the text of the manuscript. Every figure reported was produced by running the experiment code. The authors reviewed, verified and edited all output and take full responsibility for the content of this article.

8 References

- [1] S. Vosoughi, D. Roy, and S. Aral, “The spread of true and false news online,” *Science*, vol. 359, no. 6380, pp. 1146-1151, 2018.
- [2] E. Aïmeur, S. Amri, and G. Brassard, “Fake news, disinformation and misinformation in social media: a review,” *Social Network Analysis and Mining*, vol. 13, no. 30, 2023.
- [3] B. Hu, Q. Sheng, J. Cao, Y. Shi, Y. Li, D. Wang, and P. Qi, “Bad actor, good advisor: exploring the role of large language models in fake news detection,” in *Proc. AAAI*, 2024.
- [4] C. Chen and K. Shu, “Combating misinformation in the age of LLMs: opportunities and challenges,” *AI Magazine*, vol. 46, no. 2, 2024.
- [5] C. Chen and K. Shu, “Can LLM-generated misinformation be detected?” in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2024.
- [6] C. Castillo, M. Mendoza, and B. Poblete, “Information credibility on Twitter,” in *Proc. WWW*, 2011.
- [7] V. Pérez-Rosas, B. Kleinberg, A. Lefevre, and R. Mihalcea, “Automatic detection of fake news,” in *Proc. COLING*, 2018.
- [8] M. R. Kondamudi, S. R. Sahoo, L. Chouhan, and N. Yadav, “A comprehensive survey of fake news in social networks: attributes, features, and detection approaches,” *Journal of King Saud University - Computer and Information Sciences*, vol. 35, no. 6, 2023.
- [9] W. Y. Wang, “‘Liar, liar pants on fire’: a new benchmark dataset for fake news detection,” in *Proc. ACL*, 2017.
- [10] K. Pelrine, J. Danovitch, and R. Rabbany, “The surprising performance of simple baselines for misinformation detection,” in *Proc. Web Conf. (WWW)*, 2021, pp. 3432-3441.
- [11] J. Wu and B. Hooi, “Probing spurious correlations in popular event-based rumor detection benchmarks,” in *Proc. ECML-PKDD*, 2022, pp. 274-290.
- [12] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019.
- [13] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: a robustly optimized BERT pretraining approach,” arXiv:1907.11692, 2019.
- [14] T. Wolf *et al.*, “Transformers: state-of-the-art natural language processing,” in *Proc. EMNLP (System Demonstrations)*, 2020.
- [15] Z. Guo, M. Schlichtkrull, and A. Vlachos, “A survey on automated fact-checking,” *Transactions of the ACL*, vol. 10, pp. 178-206, 2022.
- [16] N. Hassan, F. Arslan, C. Li, and M. Tremayne, “Toward automated fact-checking: detecting check-worthy factual claims by ClaimBuster,” in *Proc. KDD*, 2017.
- [17] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, “FEVER: a large-scale dataset for fact extraction and verification,” in *Proc. NAACL-HLT*, 2018.
- [18] L. Derczynski, K. Bontcheva, M. Liakata, R. Procter, G. W. S. Hoi, and A. Zubiaga, “SemEval-2017 Task 8: RumourEval: determining rumour veracity and support for rumours,” in *Proc. SemEval*, 2017.
- [19] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘Why should I trust you?’ Explaining the predictions of any classifier,” in *Proc. KDD*, 2016.
- [20] K. Shu, L. Cui, S. Wang, D. Lee, and H. Liu, “dEFEND: explainable fake news detection,” in *Proc. KDD*, 2019.
- [21] X. Zhou and R. Zafarani, “A survey of fake news: fundamental theories, detection methods, and opportunities,” *ACM Computing Surveys*, vol. 53, no. 5, 2020.
- [22] A. Zubiaga, A. Aker, K. Bontcheva, M. Liakata, and R. Procter, “Detection and resolution of rumours in social media: a survey,” *ACM Computing Surveys*, vol. 51, no. 2, 2018.
- [23] H. Wan, J. Wu, M. Luo, Z. Zeng, and Z. Su, “Truth over tricks: measuring and mitigating shortcut learning in misinformation detection,” arXiv:2506.02350, 2025.
- [24] K. Shu, D. Mahudeswaran, S. Wang, D. Lee, and H. Liu, “FakeNewsNet: a data repository with news content, social context, and spatiotemporal information for studying fake news on social media,” *Big Data*, vol. 8, no. 3, pp. 171-188, 2020.

- [25] A. Silva, L. Luo, S. Karunasekera, and C. Leckie, “Embracing domain differences in fake news: cross-domain fake news detection using multi-modal data,” in *Proc. AAAI*, 2021.
- [26] A. Failla and G. Rossetti, “‘I’m in the Bluesky tonight’: insights from a year’s worth of social data,” *PLOS ONE*, vol. 19, no. 11, 2024.
- [27] A. Paszke *et al.*, “PyTorch: an imperative style, high-performance deep learning library,” in *Proc. NeurIPS*, 2019.
- [28] S. M. Hasan, S. Roy, and A. J. Nafis, “Generalization gaps in political fake news detection: an empirical study on the LIAR dataset,” arXiv:2512.18533, 2025.
- [29] Y. Dou, K. Shu, C. Xia, P. S. Yu, and L. Sun, “User preference-aware fake news detection,” in *Proc. SIGIR*, 2021.
- [30] S. Gong, R. O. Sinnott, J. Qi, and C. Paris, “Fake news detection through graph-based neural networks: a survey,” arXiv:2307.12639, 2023.
- [31] H. T. Phan, N. T. Nguyen, and D. Hwang, “Fake news detection: a survey of graph neural network methods,” *Applied Soft Computing*, vol. 139, 2023.